

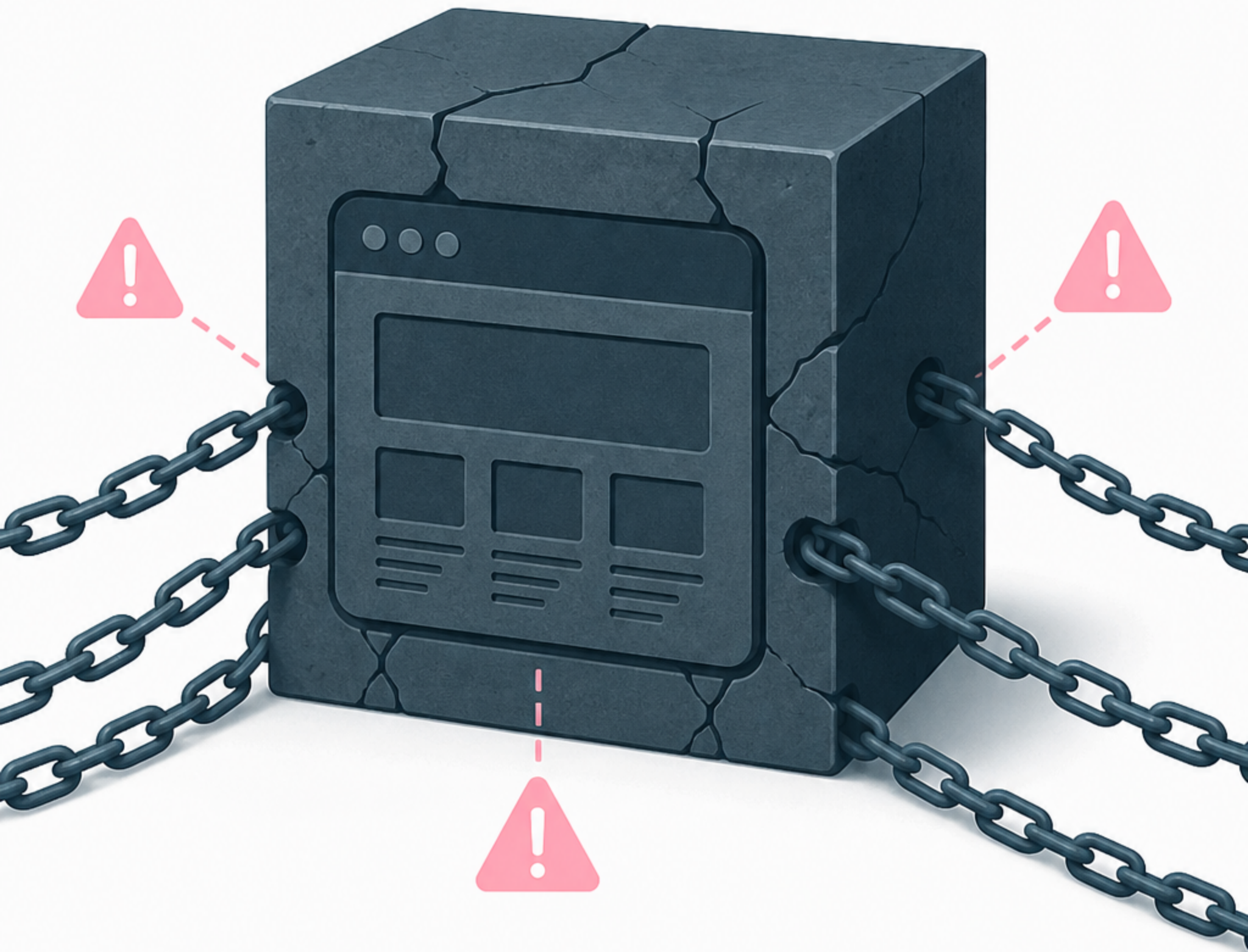
Beyond Monoliths:

Architecting Micro-Frontends
for Enterprise Scale



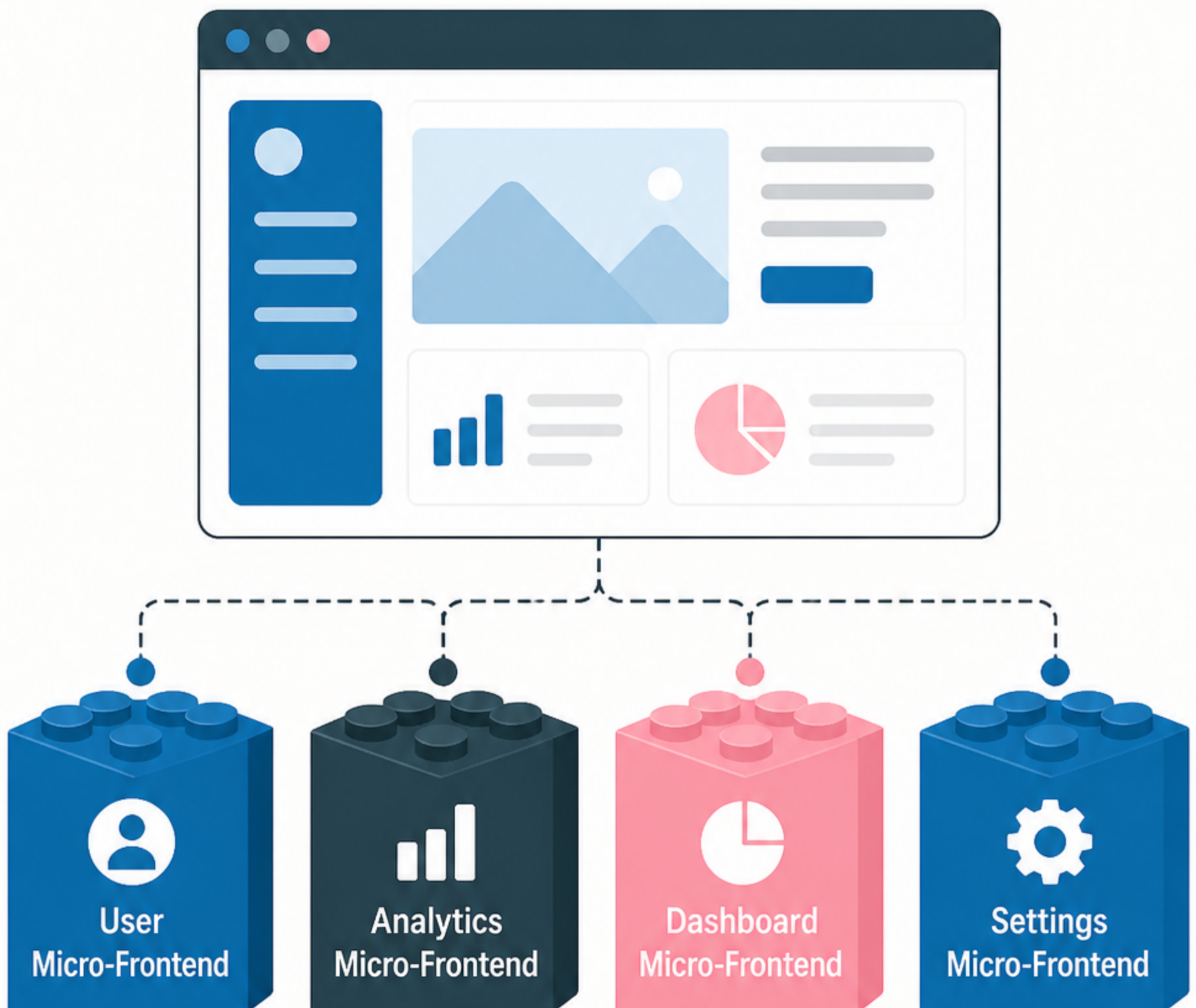
1. The Monolith Problem

Single, tightly coupled frontend apps become bottlenecks. They hinder independent deployments, slow down feature development, and scale poorly. Teams trip over each other, leading to integration hell and reduced agility.



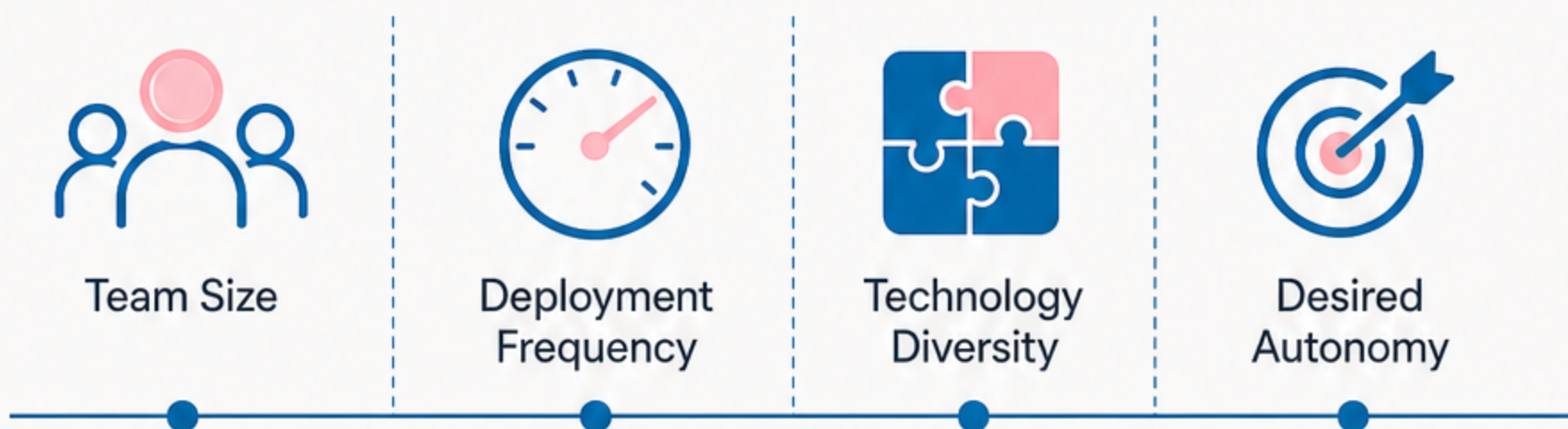
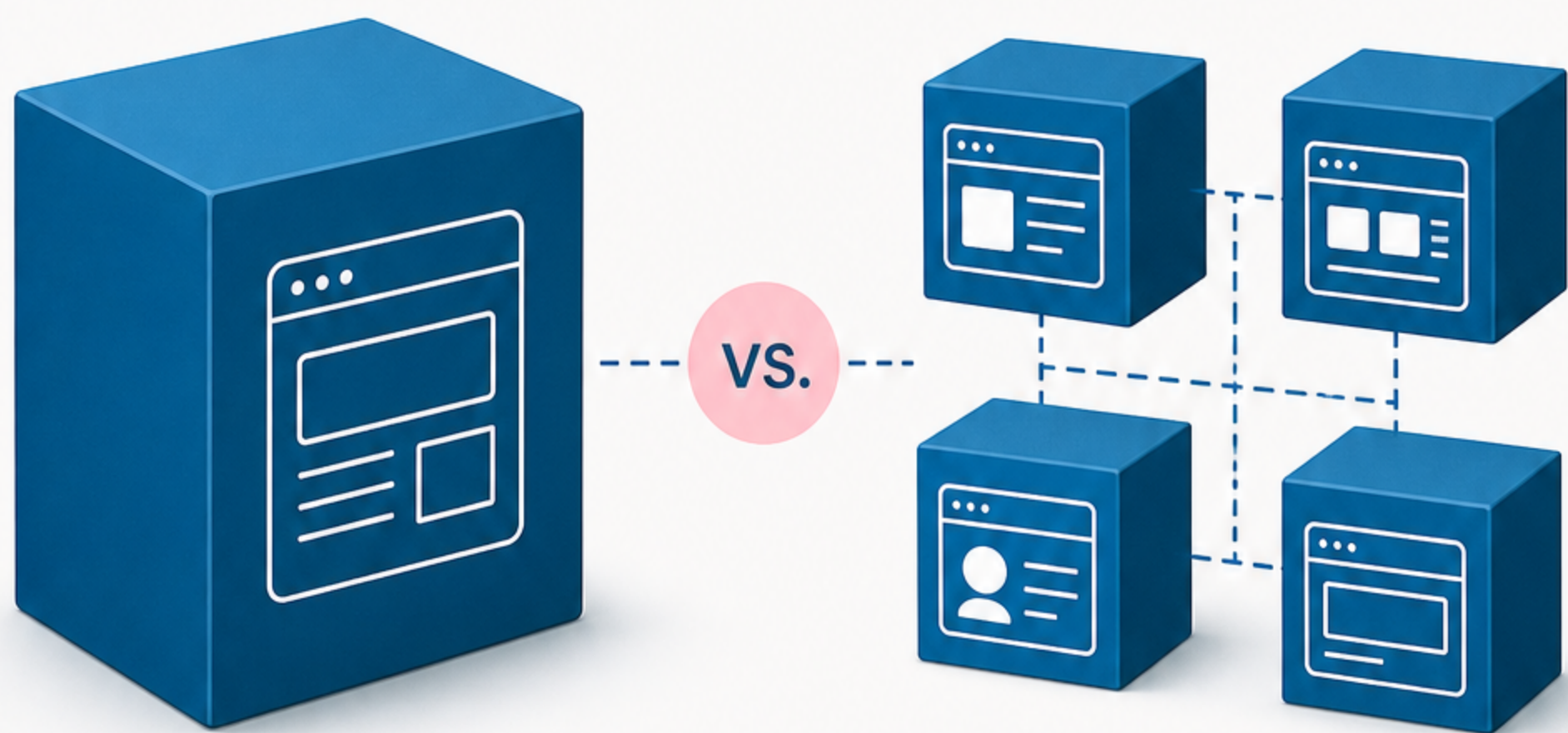
2. What are Micro-Frontends?

An architectural style where a web application is decomposed into smaller, independent 'micro-apps' that can be developed, deployed, and managed by autonomous teams. Each micro-frontend owns its domain logic and UI.



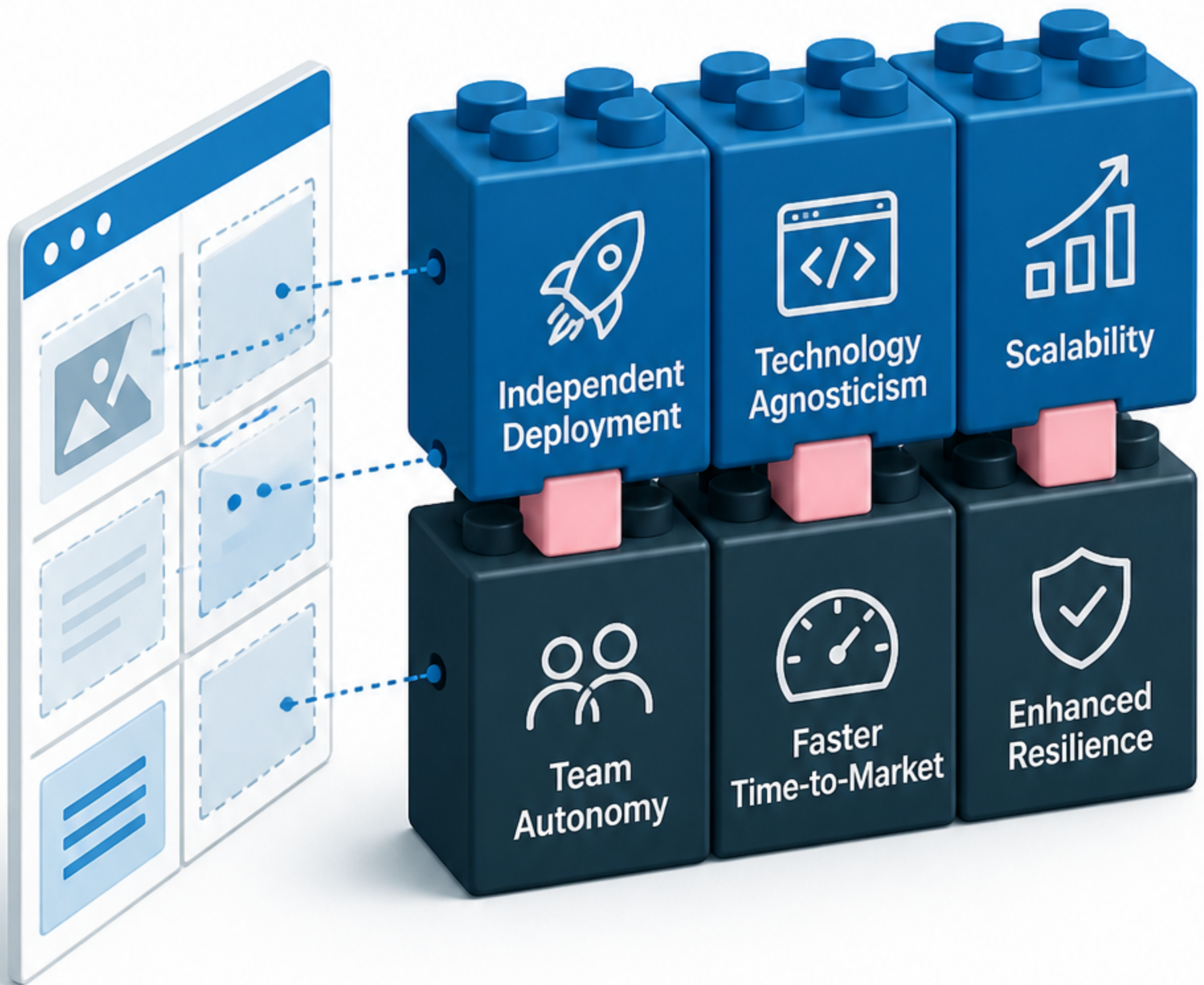
3. Monolith vs. Micro-Frontend: The Decision Matrix

Use this framework to evaluate your current architecture and future needs. Assess team size, deployment frequency, technology diversity, and desired autonomy to make an informed choice.



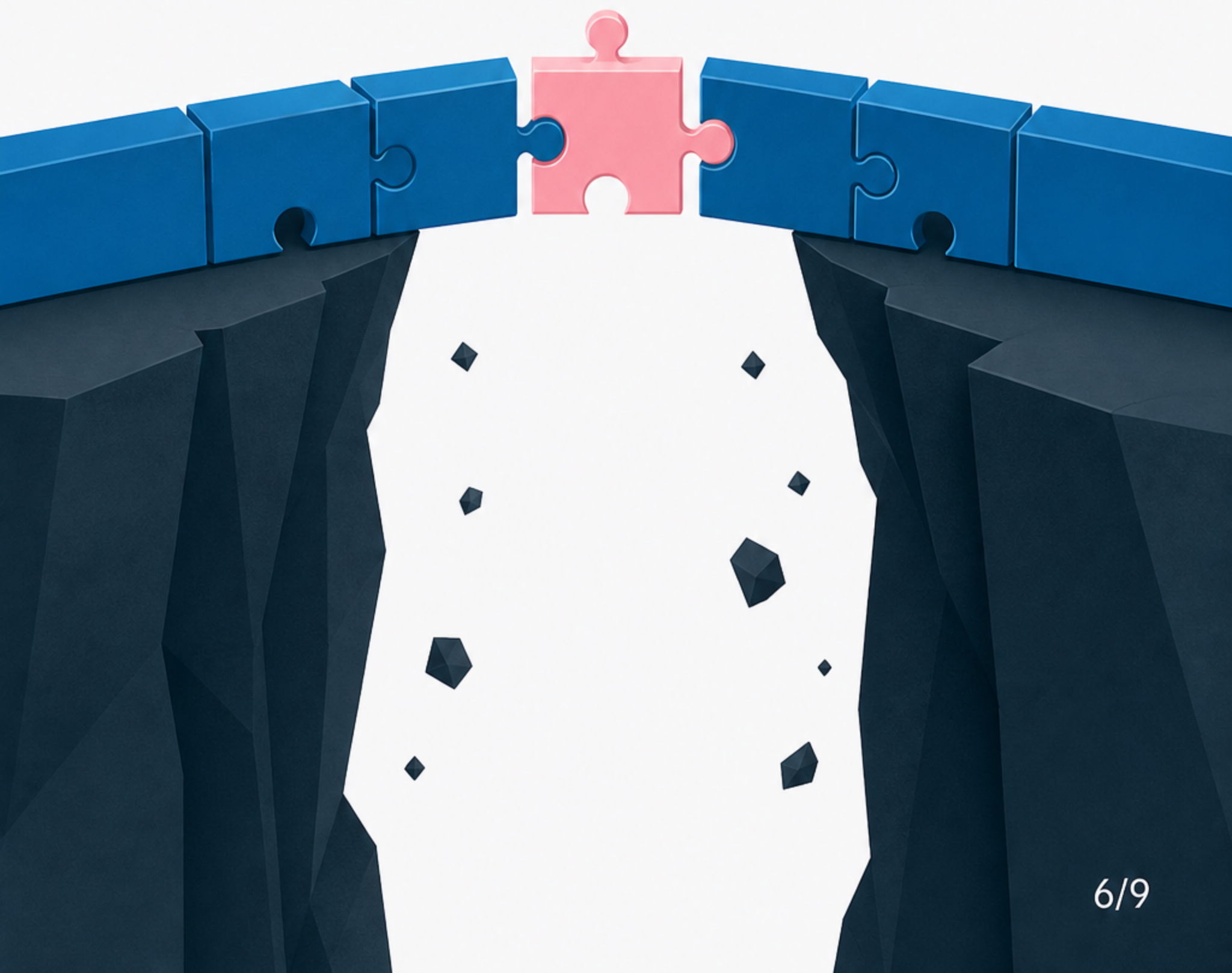
4. Key Benefits of Micro-Frontends

Independent Deployment, Technology Agnosticism, Scalability, Team Autonomy, Faster Time-to-Market, Enhanced Resilience. These benefits translate directly into business value and developer satisfaction.



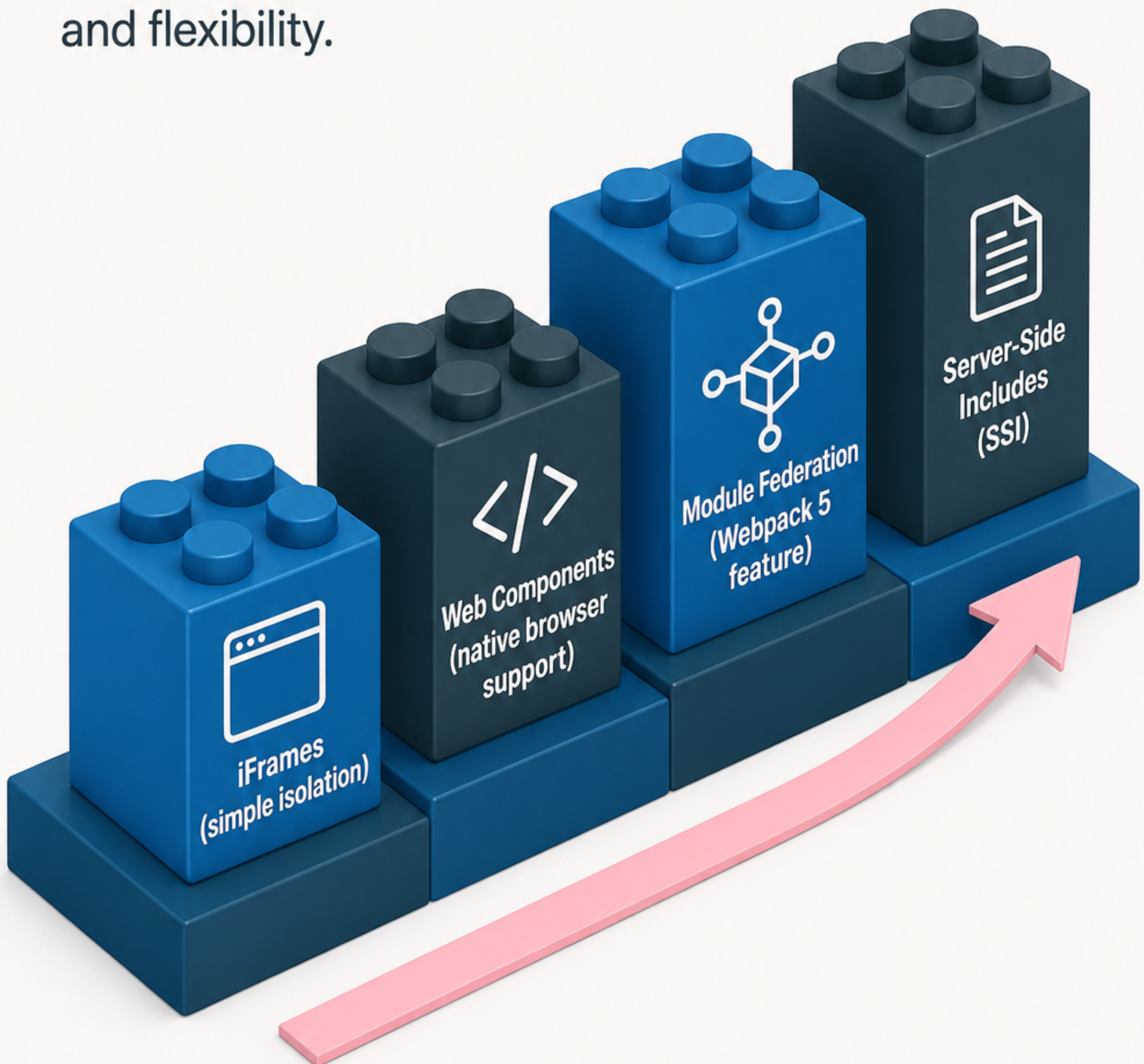
5. Common Pitfalls to Avoid

Over-fragmentation, inconsistent UX, complex communication between micro-frontends, shared state management issues, and governance overhead. Plan carefully to mitigate these challenges.



6. Implementation Strategies

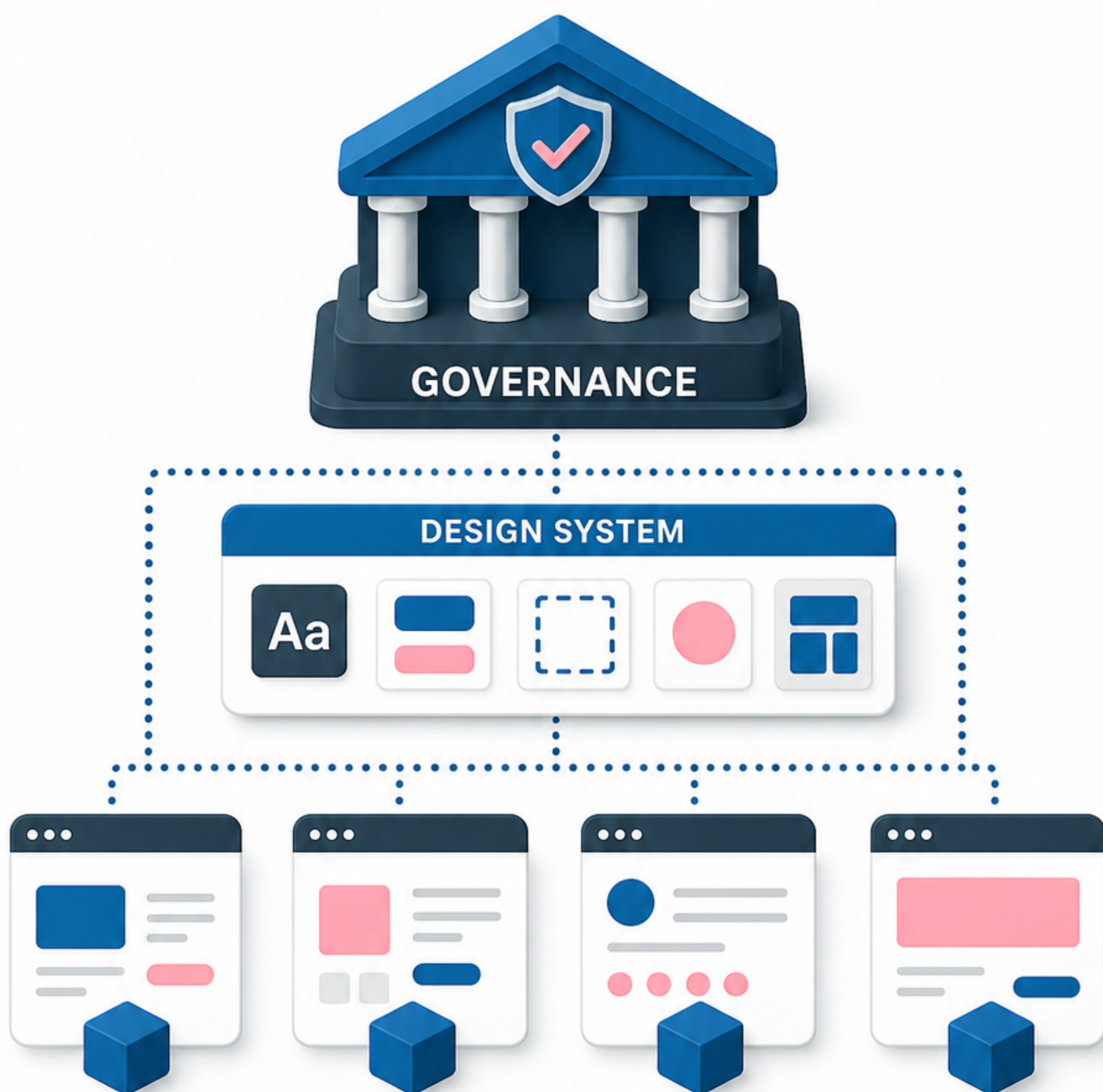
Choose wisely: iFrames (simple isolation), Web Components (native browser support), Module Federation (Webpack 5 feature), or Server-Side Includes (SSI). Each has trade-offs in complexity, performance, and flexibility.



7. Governance & Design Systems

Centralized governance for core infrastructure and a robust design system are crucial.

They ensure consistency, reduce duplication, and maintain a cohesive user experience across all micro-frontends.



What's Your Micro-Frontend Story?

Share your biggest challenge or success in adopting a micro-frontend architecture. Let's learn from each other's experiences!

